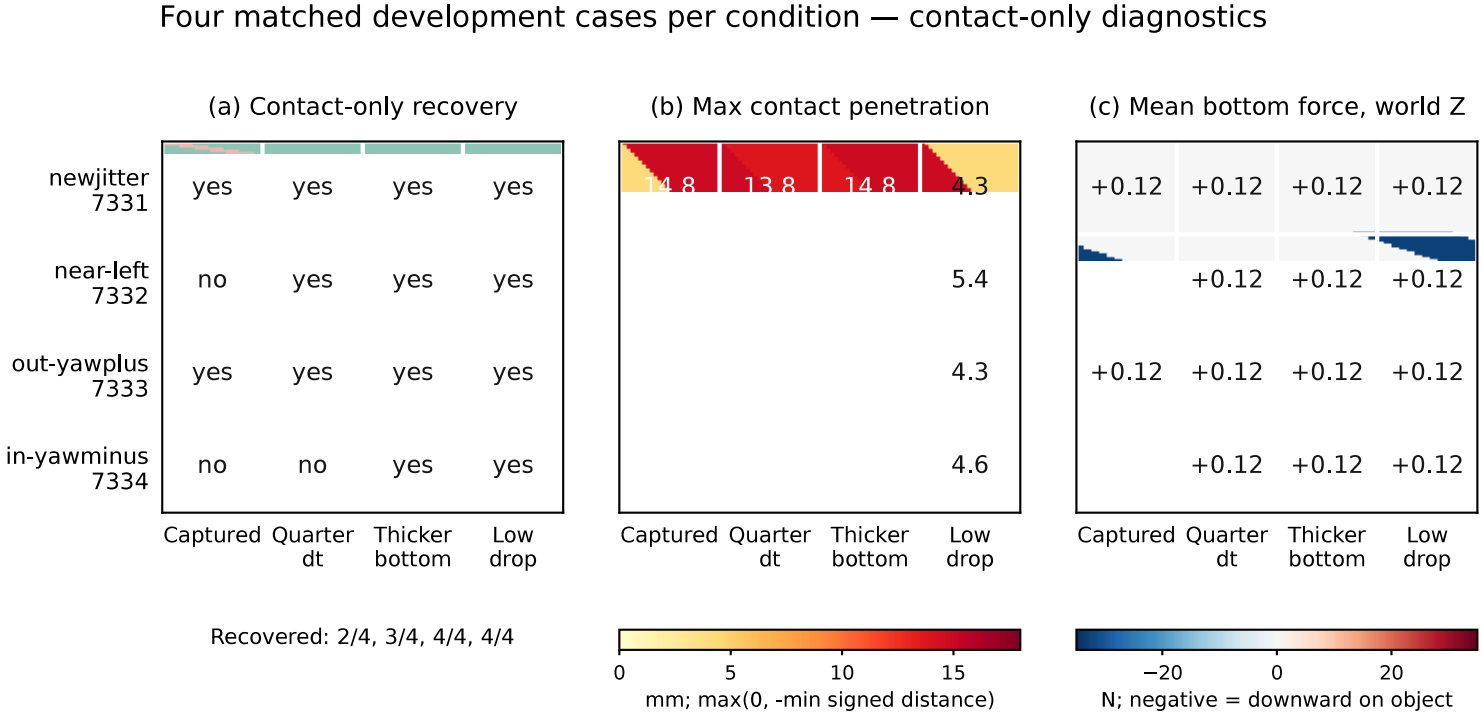

Four matched development cases per condition — contact-only diagnostics
(a) Contact-only recovery
(b) Max contact penetration
(c) Mean bottom force, world Z
newjitter 7331
near-left 7332
out-yawplus 7333
in-yawminus 7334
yes yes yes yes
no yes yes yes
yes yes yes yes
no no yes yes
14.8 13.8 14.8 4.3
5.4
4.3
4.6
+0.12 +0.12 +0.12 +0.12
+0.12 +0.12 +0.12
+0.12 +0.12 +0.12 +0.12
+0.12 +0.12 +0.12
Captured
Quarter dt
Thicker bottom
Low drop
Recovered: 2/4, 3/4, 4/4, 4/4
0 5 10 15
mm; max(0, -min signed distance)
−20 0 20
N; negative = downward on object
16 new attempts; 14,000 substeps are not independent trials. Recovery does not validate physics or a full skill.